

3 hours

Problem 2 (96RUS94)

Let $a_1, \dots, a_{1600}$ be a sequence such that a_i denotes how many committees the i -th delegate is in. Clearly, $\sum a_i = 80 \cdot 16000$.

For any formation of committees, if we were randomly to select a pair of committees, the expected number of overlaps would be,

$$\frac{\text{Total overlap}}{\binom{16000}{2}} = \frac{\sum \binom{a_i}{2}}{\binom{16000}{2}}$$

By *QM-AM*, since the sum is fixed, $\frac{\sum \binom{a_i}{2}}{\binom{16000}{2}}$ is minimized when all a_i are equal at 800. Here,

$$\frac{\sum \binom{a_i}{2}}{\binom{16000}{2}} > \frac{1600 \cdot \binom{800}{2}}{\binom{16000}{2}} > 3$$

Therefore, by Pigeonhole, there are two committees sharing at least 4.

Problem 3 (19EGMO5)

Note that the permuting $a_1, \dots, a_n$ will not affect the result since we can permute $b_1, \dots, b_n$ in the exact same way.

Let S be the multiset of remainders of $a_1, \dots, a_n$ when divided by n . We sort the elements of S to be $s_0 \leq s_1 \leq \dots \leq s_{n-1}$. We let sequence $d_0, \dots, d_{n-1}$ be a sequence where $d_i := s_i - i$ for all $0 \leq i \leq n-1$. Let $d_j = \max(d_0, \dots, d_{n-1})$ for some j .

Let $S' = \{s'_0, s'_1, \dots, s'_{n-1}\}$ be the multiset of remainders of $a_1 + (n - d_j), a_2 + (n - d_j), \dots, a_n + (n - d_j)$ when divided by n respectively. For convenience, WLOG, let $a_1, \dots, a_n$ be permuted in such a way that we already have $s'_0 \leq s'_1 \leq \dots \leq s'_{n-1}$.

Claim: $i \geq s'_i$ for all i .

Note that $s'_i = s_{i+j} - s_j$ if $i + j < n$ and $s'_i = n + s_{i+j-n} - s_j$ if $i + j \geq n$

If $i + j < n$, we know $s_{i+j} - (i + j) \leq s_j - j$ meaning $s_{i+j} - s_j \leq i$.

If $i + j \geq n$, we know $s_{i+j-n} - (i + j - n) \leq s_j - j$ meaning $n + s_{i+j-n} - s_j \leq i$ also.

If we let $b_1, \dots, b_n$ be the smallest positive integers greater than or equal to $a_1, \dots, a_n$ such that,

$$b_i \equiv (n - d_j) + (i - 1) \pmod{n}$$

Condition (a) is immediately satisfied, condition (b) is also clearly satisfied as $(n - d_j)$ is constant.

For condition (c), we have $b_i - a_i = (i - 1) - s'_{i-1}$ (indices offset due to zero vs. one indexing).

$$\begin{aligned}
b_1 + \dots + b_n &\leq n \left(\frac{n-1}{2} + \left\lfloor \frac{a_1 + \dots + a_n}{n} \right\rfloor \right) \\
\Leftrightarrow (b_1 + \dots + b_n) - (a_1 + \dots + a_n) &\leq n \left(\frac{n-1}{2} + \left\lfloor \frac{a_1 + \dots + a_n}{n} \right\rfloor \right) - (a_1 + \dots + a_n) \\
\Leftrightarrow -(s'_0 + \dots + s'_{n-1}) &\leq n \left(\left\lfloor \frac{a_1 + \dots + a_n}{n} \right\rfloor \right) - (a_1 + \dots + a_n) \\
\Leftrightarrow (s'_0 + \dots + s'_{n-1}) &\geq (a_1 + \dots + a_n) - n \left(\left\lfloor \frac{a_1 + \dots + a_n}{n} \right\rfloor \right)
\end{aligned}$$

Note that $(a_1 + \dots + a_n) - n \left(\left\lfloor \frac{a_1 + \dots + a_n}{n} \right\rfloor \right)$ would be the remainder r that $a_1 + \dots + a_n$ leaves when divided by n . Note that the remainder $(a_1 + (n - d_j)) + (a_2 + (n - d_j)) + \dots + (a_n + (n - d_j))$ leaves when divided by n is the same as r since we added $n(n - d_j)$ to the sum. Therefore, $(s'_0 + \dots + s'_{n-1})$ the sum of the individual remainders $a_1 + (n - d_j), \dots, a_n + (n - d_j)$ leaves upon division by n leaves is greater than or equal to r as desired.

8 clubs; 5 hours

Problem 4 (17BAMO4)

We plot the n -gon onto the Cartesian plane. For a point (x, y) inside the convex n -gon, let $d_1(x, y), d_2(x, y), \dots, d_n(x, y)$ be the distances from (x, y) to each of the n sides. Define $d(x, y) = \min(d_1(x, y), \dots, d_n(x, y))$. We clearly have d is continuous function and finite within the interior of the n -gon meaning there must be a point P in the interior which maximizes $d(P)$ by the extreme value theorem.

We may let $h = d(P)$ to cover the entire rectangle as all points within the rectangle are at most $d(P)$ from the nearest side. Let the side lengths of the rectangle be $s_1, \dots, s_n$. The total area of our rectangles is,

$$h \times (s_1 + \dots + s_n)$$

The total area of the n -gon would be,

$$\begin{aligned}
&\frac{1}{2}s_1d_1(P) + \frac{1}{2}s_2d_2(P) + \dots + \frac{1}{2}s_nd_n(P) \\
&\geq \frac{1}{2}(s_1 + \dots + s_n)d(P) = \frac{1}{2}h(s_1 + \dots + s_n)
\end{aligned}$$

Problem 6 (14BAMO5)

Sketch: Let $w_0, w_1, \dots, w_{2n+1}$ be the number of wins each player has. If $k = 0$, the values are respectively $0, 1, \dots, 2n$. We swap the results of a match k times. For each swap, we increment one w_i and decrement

another w_j . FTSOC, if we could have gotten $n - \sqrt{2k} > w_i$ or $w_i > n + \sqrt{2k}$, for all i , we'd need to make more than k changes.

14 clubs; 6 hours

Problem 9 (08PTNMB3)

We claim the answer is $\sqrt{\frac{n}{2}}$.

Note that if the center of the circle is not the center of hypercube, we may translate the circle to the center, where the minimum distance to sides would be greater and still have it work. Therefore, assume the circle is centered at the center of the hypercube.

Theorem: All circles of radius r exactly be parametrized as $p(t) = \vec{a} + \vec{b} \cos(t) + \vec{c} \sin(t)$ for some vectors $\vec{b}, \vec{c}$ such that $|\vec{b}| = |\vec{c}| = r$ and $\vec{b} \cdot \vec{c} = 0$.

We let $\vec{a} = \begin{pmatrix} a_1 \\ \dots \\ a_n \end{pmatrix}; \vec{b} = \begin{pmatrix} b_1 \\ \dots \\ b_n \end{pmatrix}; \vec{c} = \begin{pmatrix} c_1 \\ \dots \\ c_n \end{pmatrix}$. We see that,

$$p(t) = \begin{pmatrix} a_1 + b_1 \cos(t) + c_1 \sin(t) \\ \dots \\ a_n + b_n \cos(t) + c_n \sin(t) \end{pmatrix}$$

The circle lies within the hypercube if and only if $|a_i + b_i \cos(t) + c_i \sin(t)| \leq 1$ always for all i . To maximize $|\vec{b}| = |\vec{c}|$ and hence r , we let $a_1 = \dots = a_n = 0$. We then have,

$$|b_i \cos(t) + c_i \sin(t)| \leq 1$$

$$\Leftrightarrow \sqrt{b_i^2 + c_i^2} \leq 1$$

$$\Leftrightarrow b_i^2 + c_i^2 \leq 1$$

Summing all inequalities up,

$$\Rightarrow (b_1^2 + b_2^2 + \dots + b_n^2) + (c_1^2 + c_2^2 + \dots + c_n^2) \leq n$$

$$\Leftrightarrow 2r^2 \leq n$$

$$\Leftrightarrow r \leq \sqrt{\frac{n}{2}}$$

This proves the bound. For the construction, if n is even, we may take,

$$a_1 = \dots = a_{\frac{n}{2}} = 1; a_{\frac{n}{2}+1} = \dots = a_n = 0; b_1 = \dots = b_{\frac{n}{2}} = 0; b_{\frac{n}{2}+1} = \dots = b_n = 1$$

This gives $|\vec{b}| = |\vec{c}| = \sqrt{\frac{n}{2}}; \vec{b} \cdot \vec{c} = 0; a_i^2 + b_i^2 \leq 1$. If n is odd,

$$a_1 = \frac{\sqrt{3}}{2}; a_2 = -\frac{\sqrt{3}}{2}; a_3 = 0$$

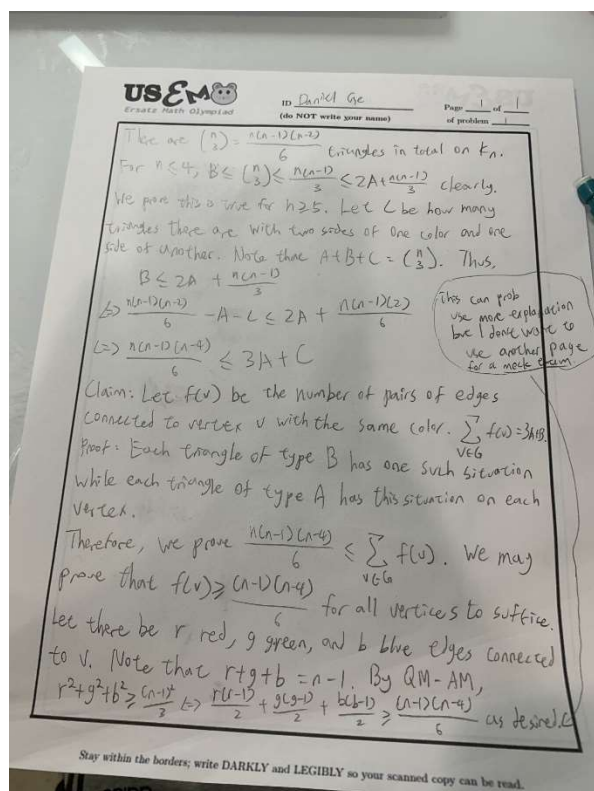
$$b_1 = b_2 = \frac{1}{2}; b_3 = 1$$

For the remaining $n - 3$ dimensions, half of them have $a_i = 1; b_i = 0$ and half $a_i = 0; b_i = 1$ as before.

We may find that $|\vec{b}| = |\vec{c}| = \sqrt{\frac{n}{2}}, \vec{b} \cdot \vec{c} = 0; a_i^2 + b_i^2 \leq 1$ as desired.

Problem 11 (23TSTST4)

(This solution was already checked by my friend after we did a mock)



20 clubs; 7 hours

Problem 12 (21SPBRG105)

We sum over the number of sides in each quadrilateral that connect two black vertices. For the sake of contradiction, if there are no quadrilaterals with a pair of adjacent black vertices and a pair of adjacent white vertices, each quadrilateral has an even number of pairs of adjacent black vertices.

Iterating over edges, there are $1 + 2 + \dots + 49 \equiv 1 \pmod{2}$ black edges of the 2550-gon. Each black diagonal counts for two to the sum. By parity, we are done.

Problem 13 (15IMO1)

- a) Using polar coordinates, we let $0, 1, \omega$ be the first three points where $\omega = \exp\left(\frac{\pi i}{3}\right)$. If n is even, we also add in ω^2 . We may then add pairs of points $a, a\omega$ for some $|a| = 1$ such that $a, a\omega$ are both new. Pairs of points on the unit circle are centered at 0 while pairs of points $0, a$ involving the origin are centered at $a\omega$ or $\frac{a}{\omega}$.
- b) The answer is exactly all odd numbers. The construction is a regular n -gon for odds. For even numbers, it is impossible by counting how many triplets (A, B, C) such that A, B are centered at C there are. There are $n(n-1)$ selections of A, B which each demand at least one C meaning at least $n(n-1)$ triplets. However, each C may host at most $\frac{n-2}{2}$ unordered pairs of A, B since it's balanced center-free for $n-2$ ordered pairs in total for $n(n-2)$ at most which is a contradiction.

26 clubs; 9 hours

Problem 15 (99RUS)

We take a random subset of the girls in the school to add to S , then add all the boys in the school who like an odd number of girls in S . Since each boy likes at least one girl, for a random selection of girls, there is a $\frac{1}{2}$ chance that he likes an odd number of those girls. Therefore, the expected size of S is the sum of the expected number of girls and expected number of boys which is half the number of boys and half the number of girls.

Noting that taking the empty set of girls will yield an empty set of boys, by Pigeonhole, there must be a selection which gives $|S|$ is greater than half the school's size.

Problem 18 (70IMO6)

Let $f(n)$ denote the maximum number of acute triangles with n points.

$$\text{Claim: } f(n+1) \leq \frac{(n+1)f(n)}{n-2}$$

Proof: We sum over all $\binom{n+1}{n}$ selections of n points to get at most $f(n)$ acute triangles with each for $(n+1)f(n)$ in total at most. Each triangle is overcounted $\binom{n-2}{n-1}$ times so $f(n+1) \leq \frac{(n+1)f(n)}{n-2}$ as desired.

Note that $f(n)$ is an integer.

$$\text{Claim: } f(n) = \left\lceil \frac{2}{3} \binom{n}{3} \right\rceil$$

This is proved by induction. For the base case, $f(4) = 3$ is obtainable by having $(-3, 0), (3, 0), (0, 1), (0, -8)$. $f(4) = 4$ is impossible. We clearly cannot have one of the points lying inside a triangle formed by the other three points. Otherwise, we may form a convex quadrilateral with the four points. One of the interior angles is not acute so one of the triangles formed is not acute.

For the inductive step, note that by casework on $n \bmod 9$,

$$\begin{aligned} f(n) &= \left\lfloor \frac{2}{3} \binom{n}{3} \right\rfloor = \frac{2}{3} \binom{n}{3} + \frac{1}{3} \left\lfloor \frac{n}{3} \right\rfloor \\ f(n+1) &\leq \frac{n+1}{n-2} \left(\frac{2}{3} \binom{n}{3} + \frac{1}{3} \left\lfloor \frac{n}{3} \right\rfloor \right) \\ &= \frac{2}{3} \binom{n+1}{3} + \frac{n+1}{n-2} \left(\frac{1}{3} \left\lfloor \frac{n}{3} \right\rfloor \right) \end{aligned}$$

To show that $\left\lfloor \frac{2}{3} \binom{n+1}{3} + \frac{n+1}{n-2} \left(\frac{1}{3} \left\lfloor \frac{n}{3} \right\rfloor \right) \right\rfloor = \frac{2}{3} \binom{n+1}{3} + \frac{1}{3} \left\lfloor \frac{n+1}{3} \right\rfloor$ to finish. We may show that,

$$\begin{aligned} \frac{2}{3} \binom{n+1}{3} + \frac{n+1}{n-2} \left(\frac{1}{3} \left\lfloor \frac{n}{3} \right\rfloor \right) &< \frac{2}{3} \binom{n+1}{3} + \frac{1}{3} \left\lfloor \frac{n+1}{3} \right\rfloor + 1 \\ &\Leftrightarrow \frac{n+1}{n-2} \left(\frac{1}{3} \left\lfloor \frac{n}{3} \right\rfloor \right) < \frac{1}{3} \left\lfloor \frac{n+10}{3} \right\rfloor \\ &\Leftrightarrow \frac{n+1}{n-2} \left(\frac{1}{3} \binom{n}{3} \right) < \frac{1}{3} \binom{n+8}{3} \\ &\Leftrightarrow \frac{n^2 + n}{n-2} < n+8 \\ &\Leftrightarrow n^2 + n < n^2 + 6n - 16 \end{aligned}$$

This is true for $n \geq 4$.

40 clubs; 11.5 hours

Problem 20 (10SLC5)

Note that,

$$\begin{aligned} &\sum \binom{w_i}{3} - \sum \binom{l_i}{3} - \sum l_i \binom{w_i}{2} + \sum w_i \binom{l_i}{2} \\ &= \frac{w_i^3 - 3w_i^2 + 2w_i}{6} - \frac{l_i^3 - 3l_i^2 + 2l_i}{6} - \frac{l_i w_i^2 - l_i w_i}{2} + \frac{l_i^2 w_i - l_i w_i}{2} \\ &= \frac{(w_i - l_i)^3}{6} + \frac{w_i - l_i}{3} + \frac{l_i^2 - w_i^2}{2} \\ &= \frac{(w_i - l_i)^3}{6} + \frac{(w_i - l_i)}{6} \cdot (2 + 3(n-1)) \end{aligned}$$

Note that $\sum w_i - l_i = 0$ so,

$$6 \left(\sum \binom{w_i}{3} - \sum \binom{l_i}{3} - \sum l_i \binom{w_i}{2} + \sum w_i \binom{l_i}{2} \right) = (w_i - l_i)^3$$

Let the signature of a group of four players in the tournament be the set of the number of wins the players scored playing against the other three. Let a be the number of groups of four from the tournament with signature $\{0, 1, 2, 3\}$, b be that of $\{1, 1, 1, 3\}$, c be that of $\{1, 1, 2, 2\}$, and d be that of $\{0, 2, 2, 2\}$.

$$\text{Claim 1: } \sum \binom{w_i}{3} = a + b; \sum \binom{l_i}{3} = a + d$$

Proof: We double count. For a given player p_a with w_i wins, for every selection of a triplet of players (p_b, p_c, p_d) they've won against, we have a group $(p_a, \dots, p_d)$ that is either a type- a or type- b . The same thing can be said for $\binom{l_i}{3} = a + d$.

$$\text{Claim 2: } \sum l_i \binom{w_i}{2} = a + 2c + 3d; \sum w_i \binom{l_i}{2} = a + 2c + 3b$$

We count how many quadruplets of players (p_a, p_b, p_c, p_d) there are where p_a loses to p_b and wins against p_c, p_d . Each type- a contributes 1, each type c contributes 2, and type d contributes 3. Same thing for the opposite.

Therefore,

$$(w_i - l_i)^3 = 6((a + b) - (a + d) - (a + 2c + 3d) + (a + 2c + 3b)) = 24b - 24d$$

Since $d = 0$, we are done.

Problem 21 (20AMO2)

The answer is 3030 which will be constructed and proved later in the writeup.

Let the six faces of the cube be called $a_0, b_0, c_0, a_{2020}, b_{2020}, c_{2020}$ where a_0, a_{2020} are parallel and so on. Let plane $a_1, a_2, \dots, a_{2019}$ be parallel to a_0, a_{2020} spaced one unit apart from a_0 to a_{2020} . Similarly define $b_1, \dots, b_{2019}$ and $c_1, \dots, c_{2019}$.

For each beam inside the triangle, we define its "signature". We start with set of planes that its four rectangular faces are a part of. For example, this can be $\{a_{1434}, a_{1435}, b_{855}, b_{85}\}$. Note that pairs of planes in the signature are adjacent, so we may simplify this further by only keeping the one with the higher number. The signature would then become the unordered pair (a_{1435}, b_{856}) . We also define its orientation to be the letter of the plane its 1×1 faces touch, which is c in this case.

For no two beams to have intersecting interiors if and only if their orientations and signatures are the same or if one of the elements of their signature is the same and they have different orientations. For example, (a_{1435}, b_{856}) overlaps with both itself and (a_{1435}, b_{1434}) .

For convenience, let the signatures of the faces of the cube be $a_0, a_{2021}, b_0, b_{2021}, c_0, c_{2021}$. A beam (a_n, b_m) satisfies each of its interior faces touching another face if and only if, for each of

$a_{n-1}, a_{n+1}, b_{m-1}, b_{m+1}$, it is either present in the signature another beam with different orientation or present in a beam with the same orientation that is one space apart from our beam.

Claim: All $a_0, \dots, a_{2021}, b_0, \dots, b_{2021}$, and $c_0, \dots, c_{2021}$ are present in some signatures if we have less than 2020^2 beams.

Proof: We select a beam from the configuration at random. WLOG, have it orientation c and signature (a_n, b_m) . Note that we need a beam with a_{n-1}, a_{n+1} in its signature, which then requires a_{n-2}, a_{n+2} and so on inducting to the edge. Same thing can be said about b_m . If we have a beam of orientation a or b then $c_0, \dots, c_{2021}$ are all required for the same reason. If all our beams have orientation c , then we clearly must fill up the whole cube with 2020^2 beams.

Therefore, it is impossible to get under 3030 unless if $a_0, \dots, a_{2021}, b_0, \dots, b_{2021}$, and $c_0, \dots, c_{2021}$ are all present. We know that $a_0, a_{2021}, b_0, b_{2021}, c_0, c_{2021}$ are the faces. For the other 6060 signature elements, we need at least $\frac{6060}{2} = 3030$ beams to include them all.

Finally, for the construction, we take the beams,

$$\begin{aligned} & (a_1, b_2), (b_1, c_2), (c_1, a_2), \\ & (a_3, b_4), (b_3, c_4), (c_3, a_4), \\ & \dots \\ & (a_{2019}, b_{2020}), (b_{2019}, c_{2020}), (c_{2019}, a_{2020}) \end{aligned}$$

Problem 22 (10AMO6)

The answer is 43.

We first prove that we can always attain 43. For each non-negative integer n less than or equal to the greatest absolute value of any of the 136 integers, we flip a biased coin to determine which of n or $-n$ to add to S . For each integer x in S , we remove all occurrences of x among the 136 integers.

For the coin flip for n , if (n, n) is one of the ordered pairs, then we have $\frac{\sqrt{5}-1}{2}$ chance for positive n . Otherwise, we give $\frac{\sqrt{5}-1}{2}$ chance for negative n . We count the expected value of our score. For each ordered pair of the form (n, n) , we have a $\frac{\sqrt{5}-1}{2}$ chance that n is erased. For each ordered pair of the form (m, n) for $m \neq n$, we have at least a $\frac{3-\sqrt{5}}{2}$ chance that m, n are erased respectively with independent probabilities. The chance that at least one is erased is $\frac{3-\sqrt{5}}{2} + \frac{3-\sqrt{5}}{2} - \left(\frac{3-\sqrt{5}}{2}\right)\left(\frac{3-\sqrt{5}}{2}\right) = \frac{\sqrt{5}-1}{2}$ as well. Therefore, the total expected value is $68\left(\frac{\sqrt{5}-1}{2}\right)$. We see that,

$$68\left(\frac{\sqrt{5}-1}{2}\right) > 42 \Leftrightarrow 34\sqrt{5} > 76 \Leftrightarrow 5780 > 5776$$

This means there is an outcome where the score is at least 43 meaning there exists a construction.

We now prove that 43 is the limit. We take $(1, 1), (2, 2), \dots, (8, 8)$ each 5 times. We then take all pairs (a, b) for $-1 \geq a > b \geq -8$. In total, we have 68 ordered pairs. If we take p positives and $8 - p$ negatives to put in S , we'd have,

$$5p + \left(7 \cdot (8 - p) - \binom{8 - p}{2} \right)$$

We may check that this is capped at 43.

59 clubs; 17 hours